

BOIZENBURG, Germany

WMO: 102490

Lat: 53.3911N

Lon: 10.6878E

Elev: 151

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.7	14.6	2.8	6.4	11.7	7.5	8.1	15.2	29.2	42.3	25.6	43.2	6.0	30	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	83.3	65.3	79.9	64.1	76.2	62.4	67.1	78.9	65.4	76.6	63.6	74.0	9.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.0	86.6	71.9	61.0	80.6	70.1	59.0	75.0	68.2	31.6	79.4	30.3	76.8	29.0	73.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	20.3	18.1	8.3	89.2	10.0	4.3	1.1	92.3	-4.7	94.8	-10.3	97.2	-17.6	100.3
			7.4	71.5	9.7	1.5	0.4	72.6	-5.2	73.5	-10.7	74.3	-17.7	75.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	32.6	32.9	39.4	45.6	54.0	59.4	63.3	63.1	56.8	49.2	40.1	36.0
	DBStd	12.91	10.34	8.60	6.68	5.92	6.55	6.07	5.43	5.19	4.59	6.35	6.04	6.96
	HDD50	2368	539	481	333	155	33	1	0	0	3	89	299	435
	HDD65	6396	1004	899	795	581	345	188	98	101	249	489	747	900
	CDD50	1566	0	1	3	25	156	285	413	407	208	66	2	0
	CDD65	118	0	0	0	0	4	22	46	42	4	0	0	0
	CDH74	1170	0	0	0	1	65	232	488	352	31	1	0	0
	CDH80	255	0	0	0	0	5	48	123	77	2	0	0	0
Wind	WSAvg	8.9	10.7	9.9	9.9	8.4	8.0	7.4	7.7	7.3	8.4	9.0	9.5	10.3
Precipitation	PrecAvg	28.0	2.5	2.0	1.9	1.6	2.1	2.7	3.1	3.0	2.2	2.1	2.3	2.5
	PrecMax	37.4	4.8	4.9	4.5	3.5	4.3	5.9	6.9	7.2	6.5	5.6	4.6	5.9
	PrecMin	20.5	0.1	0.3	0.2	0.1	0.4	0.6	0.4	1.0	0.5	1.0	0.2	1.1
	PrecStd	4.5	1.3	1.0	1.1	0.8	1.0	1.3	1.6	1.5	1.3	1.0	1.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.1	57.7	62.3	70.2	80.1	86.0	87.6	86.3	78.2	70.6	56.0	53.3
		MCWB	47.9	51.0	51.2	56.2	62.0	66.8	65.8	66.6	64.6	62.4	50.7	49.2
	2%	DB	48.7	51.4	55.7	65.0	75.4	80.9	83.5	82.4	72.5	64.7	53.2	50.3
		MCWB	45.2	46.3	47.9	53.0	59.5	64.7	65.6	65.0	61.8	57.2	49.4	47.4
	5%	DB	46.5	46.8	51.5	60.6	71.7	76.2	80.0	78.3	68.2	60.9	51.0	48.0
		MCWB	43.0	42.8	45.5	50.2	58.1	61.7	64.6	63.7	58.9	54.8	47.7	45.2
	10%	DB	44.3	43.3	48.7	56.5	67.3	71.9	75.7	74.1	65.0	58.1	48.6	45.9
		MCWB	41.2	39.7	44.0	48.2	56.4	59.9	62.8	62.1	57.1	53.0	45.9	43.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.3	51.3	51.9	57.7	63.5	69.5	70.2	68.9	66.0	62.8	52.4	50.3
		MCDB	51.1	57.2	60.7	67.8	76.9	80.6	82.4	81.6	74.8	70.1	54.5	52.7
	2%	WB	45.7	46.3	48.8	53.9	61.1	66.2	67.3	66.3	62.7	58.2	49.7	47.8
		MCDB	48.5	50.5	54.4	62.5	72.6	77.2	79.5	77.9	70.2	63.3	52.4	50.0
	5%	WB	43.5	42.9	46.5	51.3	58.9	63.4	65.3	64.7	60.5	55.5	48.0	45.5
		MCDB	46.2	46.6	50.8	58.6	69.5	72.5	77.7	75.4	66.7	60.0	50.5	47.8
	10%	WB	41.3	40.0	44.2	49.2	56.6	60.9	63.2	63.0	58.3	53.4	46.0	43.4
		MCDB	44.1	43.0	48.2	55.9	66.3	70.0	74.5	72.7	63.6	57.5	48.8	45.8
Mean Daily Temperature Range	5% DB	MDBR	8.2	9.9	11.8	15.6	17.5	16.8	17.7	17.1	14.4	12.2	8.4	7.2
		MCDBR	8.9	12.7	17.5	23.0	24.9	25.0	25.6	24.6	19.6	16.2	11.3	9.3
		MCWBR	8.1	10.2	12.0	13.9	13.4	12.2	12.2	11.8	11.5	11.1	9.6	8.6
	5% WB	MCDBR	8.6	12.5	15.9	20.2	22.6	21.8	23.4	20.6	17.5	15.1	10.8	9.4
		MCWBR	8.2	10.5	11.6	13.2	13.0	11.7	12.4	10.9	11.5	10.9	9.4	8.9
Clear-Sky Solar Irradiance	taub	0.317	0.337	0.369	0.410	0.406	0.403	0.422	0.415	0.385	0.356	0.333	0.308	
	taud	2.346	2.344	2.280	2.211	2.265	2.294	2.257	2.290	2.361	2.420	2.393	2.340	
	Ebn at Noon	207	240	256	259	267	269	261	255	249	231	199	186	
	Edh at Noon	19	25	33	40	40	39	40	37	30	23	18	16	
All-Sky Solar Radiation	RadAvg	185	402	772	1270	1592	1683	1582	1318	967	534	229	139	
	RadStd	22	55	75	173	200	129	228	103	109	65	24	9	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-259	N/A	N/A

Nomenclature: See separate page